

Mandon Pathak

Curriculum Vitae (Prepared April 2026)

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Research Interests

- Harmonic Analysis
 - Time-Frequency Analysis
- Geometric Measure theory
- Convex integration

Harmonic analysis, geometric measure theory and its interplay with partial differential equations.

Employment

Aug 2023– **Purdue University**, *West Lafayette, IN, USA*.
Doctoral Student, Teaching assistant/ Research assistant.

Education

Aug 2023– **Purdue University**, *West Lafayette, IN, USA*.
Ph.D. Student in Mathematics. Expected graduation Spring 2028.
Advisor: **Prof. Victor D. Lie**

2018–2023 **Indian Institute of Science Education
and Research (IISER)**, *Bhopal, MP, India*.

Dual BS-MS degree in Mathematics

Advisor: **Dr. Sombuddha Bhattacharya (IISERB)** and **Dr. Venky Krishnan (TIFR-CAM)**

Dissertation: **Partial differential equations and applications to imaging**

CPI: 9.19/10.

Ongoing Work

- **On the curved Bilinear Hilbert–Carleson Operator.**

Publication

- **Non-unique solutions to the periodic gKdV equation (with N. Gismondi, K. Ma and A. Radu).**
[arXiv:2606.06916](#)
- **Nontrivial Weak Solutions of the Stationary KdV equation in Sharp L^p Spaces.**
[arXiv:2603.12555](#).

Honors and Awards

- 2023 Recipient of the Mathematics Proficiency Gold Medal for best academic performance in the graduating batch (2023) at IISER Bhopal.
- 2017 Inspire fellowship, sponsored by the Department of Science and Technology (DST), Government of India, for being in the top 1% of India's ISC board. (not availed)

Teaching

Purdue University

- Spring 2026 MA 511 - Grader for Linear Algebra and Its Application
- Fall 2025 MA 261 - TA for Multivariable Calculus
- Spring 2025 MA 261 - TA for Multivariable Calculus
- Fall 2024 MA 261 - TA for Multivariable Calculus
- Spring 2024 MA 366 - Grader for Ordinary Differential Equations
- Fall 2023 MA 161 - TA for Plane Analytic Geometry and Calculus I

Conferences and Workshop

- May 2026 **Simons School on Discrete harmonic analysis and analytic number theory**
Alfréd Rényi Institute of Mathematics, Budapest Hungary.

References

- Dr. Victor Lie, Purdue University, vlie@purdue.edu.
- Dr. Plamen Stefanov, Purdue University, stefanov@math.purdue.edu.
- Dr. Sombuddha Bhattacharyya, IISER Bhopal, sombuddha@iiserb.ac.in.
- Dr. Venky Krishnan, TIFR-CAM, vkrishnan@tifrbng.res.in.